Assignment 12 — Memory Management

Subject: Operating Systems
Assignment No.: 12
Title: Memory Management Techniques
Difficulty: Intermediate
Submission Format: PDF / DOCX

Objective

To understand how an operating system manages main memory and provides memory to processes.

Questions

1. What is memory management?
2. Why is memory management important?
3. Explain primary memory.
4. Explain secondary memory.
5. What is memory allocation?
6. Explain contiguous memory allocation.
7. What is paging?
8. Explain pages and frames.
9. Explain segmentation.
10. Differentiate between paging and segmentation.
11. What is virtual memory?
12. Explain demand paging.
13. What is a page fault?
14. Explain internal fragmentation.
15. Explain external fragmentation.
16. List the advantages of virtual memory.
17. Explain the role of memory management in an operating system.